# A VANISHING RANGE AND AN S2 STATEMENT FOR THE RELATIVE DUALIZING COMPLEX OVER A COHEN-MACAULAY BASE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The base change of the relative dualizing complex along a localization of the base. Invoked at the start of the counted proof and again in the middle (source lines 5366-5367 and 5383-5384). Reduces to a Cohen-Macaulay local base.
- The Cohen-Macaulayness of the completion of a Cohen-Macaulay local ring. Invoked inside the counted proof (source lines 5373-5375). Allows replacing the base by its completion.
- The existence of a dualizing complex after passing to a suitable local base. Invoked inside the counted proof (source lines 5385-5388). Makes a dualizing complex available for the base.
- The normalized form of the dualizing complex of a Cohen-Macaulay local ring. Invoked inside the counted proof (source lines 5399-5402). Writes the base dualizing complex as a module in a single degree.
- The degree confinement and the depth statement for a normalized dualizing complex. Invoked inside the counted proof (source lines 5416-5418). Gives the vanishing range of the cohomology of the shriek of the base dualizing complex.

*Source reference.* Original reference algebra-lemma-flat-going-down: sources/stacks-dualizing/pinned-original-source.tex, source lines 5379–5379. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-flat-going-down, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged

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*Source reference.* Original lemma lemma-apply-CM: sources/stacks-dualizing/pinned-original-source.tex, source lines 3842–3842. Navigation pointer only: the counted text refers to the Stacks item lemma-apply-CM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-base-change-relative-algebraic: sources/stacks-dualizing/pinned-original-source.tex, source lines 5023–5023. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-relative-algebraic, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-depth-dualizing-module: sources/stacks-dualizing/pinned-original-source.tex, source lines 3576–3576. Navigation pointer only: the counted text refers to the Stacks item lemma-depth-dualizing-module, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-relative-dualizing-if-have-omega: sources/stacks-dualizing/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted text refers to the Stacks item lemma-relative-dualizing-if-have-omega, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-shriek-normalized: sources/stacks-dualizing/pinned-original-source.tex, source lines 5221–5221. Navigation pointer only: the counted text refers to the Stacks item lemma-shriek-normalized, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-sitting-in-degrees: sources/stacks-dualizing/pinned-original-source.tex, source lines 3232–3232. Navigation pointer only: the counted text refers to the Stacks item lemma-sitting-in-degrees, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original lemma lemma-ubiquity-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 4343–4343. Navigation pointer only: the counted text refers to the Stacks item lemma-ubiquity-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original reference more-algebra-lemma-completion-CM: sources/stacks-dualizing/pinned-original-source.tex, source lines 5375–5375. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-completion-CM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's

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*Source reference.* Original reference more-algebra-lemma-hom-into-S2: sources/stacks-dualizing/pinned-original-source.tex, source lines 5429–5429. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-hom-into-S2, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

*Source reference.* Original reference morphisms-lemma-dimension-fibre-after-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 5382–5382. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-dimension-fibre-after-base-change, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

*Source reference.* Original reference morphisms-lemma-dimension-formula-general: sources/stacks-dualizing/pinned-original-source.tex, source lines 5410–5410. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-dimension-formula-general, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

**Lemma 0.1.** *Let $R \to A$ be a finite type homomorphism of Noetherian rings. Let $\mathfrak{p} \subset R$ be a prime ideal. Assume*

(1) *$R_\mathfrak{p}$ is Cohen-Macaulay, and*
(2) *for any minimal prime $\mathfrak{q} \subset A$ we have $trdeg_{\kappa(R\cap\mathfrak{q})}\kappa(\mathfrak{q}) \leq r$.*

*Then*

$$H^i(\omega^\bullet_{A/R})_\mathfrak{p} \neq 0 \Rightarrow -r \leq i$$

*and $H^{-r}(\omega^\bullet_{A/R})_\mathfrak{p}$ is $(S_2)$ as an $A_\mathfrak{p}$-module.*

*Proof.* We may replace $R$ by $R_\mathfrak{p}$ by Lemma lemma-base-change-relative-algebraic. Thus we may assume $R$ is a Cohen-Macaulay local ring and we have to show the assertions of the lemma for the $A$-modules $H^i(\omega^\bullet_{A/R})$.

Let $R^\wedge$ be the completion of $R$. The map $R \to R^\wedge$ is flat and $R^\wedge$ is Cohen-Macaulay (More on Algebra, Lemma more-algebra-lemma-completion-CM). Observe that the minimal primes of $A \otimes_R R^\wedge$ lie over minimal primes of $A$ by the flatness of $A \to A \otimes_R R^\wedge$ (and going down for flatness, see Algebra, Lemma algebra-lemma-flat-going-down). Thus condition (2) holds for the finite type ring map $R^\wedge \to A \otimes_R R^\wedge$ by Morphisms, Lemma morphisms-lemma-dimension-fibre-after-base-change. Appealing to Lemma lemma-base-change-relative-algebraic once again it suffices to prove the lemma for $R^\wedge \to A \otimes_R R^\wedge$. In this way, using Lemma lemma-ubiquity-dualizing, we may assume $R$ is a Noetherian local Cohen-Macaulay ring which has a dualizing complex $\omega^\bullet_R$.

Let $\mathfrak{m} \subset A$ be a maximal ideal. It suffices to show that the assertions of the lemma hold for $H^i(\omega^\bullet_{A/R})_\mathfrak{m}$. If $\mathfrak{m}$ does not lie over the maximal ideal of $R$, then we replace $R$ by a localization to reduce to this case (small detail omitted).

We may assume $\omega_R^\bullet$ is normalized. Setting $d = \dim(R)$ we see that $\omega_R^\bullet = \omega_R[d]$ for some $R$-module $\omega_R$, see Lemma lemma-apply-CM. Set $\omega_A^\bullet = \varphi^!(\omega_R^\bullet)$. By Lemma lemma-relative-dualizing-if-have-omega we have

$$\omega_{A/R}^\bullet = R\operatorname{Hom}_A(\omega_R[d] \otimes_R^{\mathbf{L}} A, \omega_A^\bullet)$$

By the dimension formula we have $\dim(A_{\mathfrak{m}}) \leq d + r$, see Morphisms, Lemma morphisms-lemma-dimension-formula-general and use that $\kappa(\mathfrak{m})$ is finite over the residue field of $R$ by the Hilbert Nullstellensatz. By Lemma lemma-shriek-normalized we see that $(\omega_A^\bullet)_{\mathfrak{m}}$ is a normalized dualizing complex for $A_{\mathfrak{m}}$. Hence $H^i((\omega_A^\bullet)_{\mathfrak{m}})$ is nonzero only for $-d - r \leq i \leq 0$, see Lemma lemma-sitting-in-degrees. Since $\omega_R[d] \otimes_R^{\mathbf{L}} A$ lives in degrees $\leq -d$ we conclude the vanishing holds. Finally, we also see that

$$H^{-r}(\omega_{A/R}^\bullet)_{\mathfrak{m}} = \operatorname{Hom}_A(\omega_R \otimes_R A, H^{-d-r}(\omega_A^\bullet))_{\mathfrak{m}}$$

Since $H^{-d-r}(\omega_A^\bullet)_{\mathfrak{m}}$ is $(S_2)$ by Lemma lemma-depth-dualizing-module we find that the final statement is true by More on Algebra, Lemma more-algebra-lemma-hom-into-S2. $\square$